

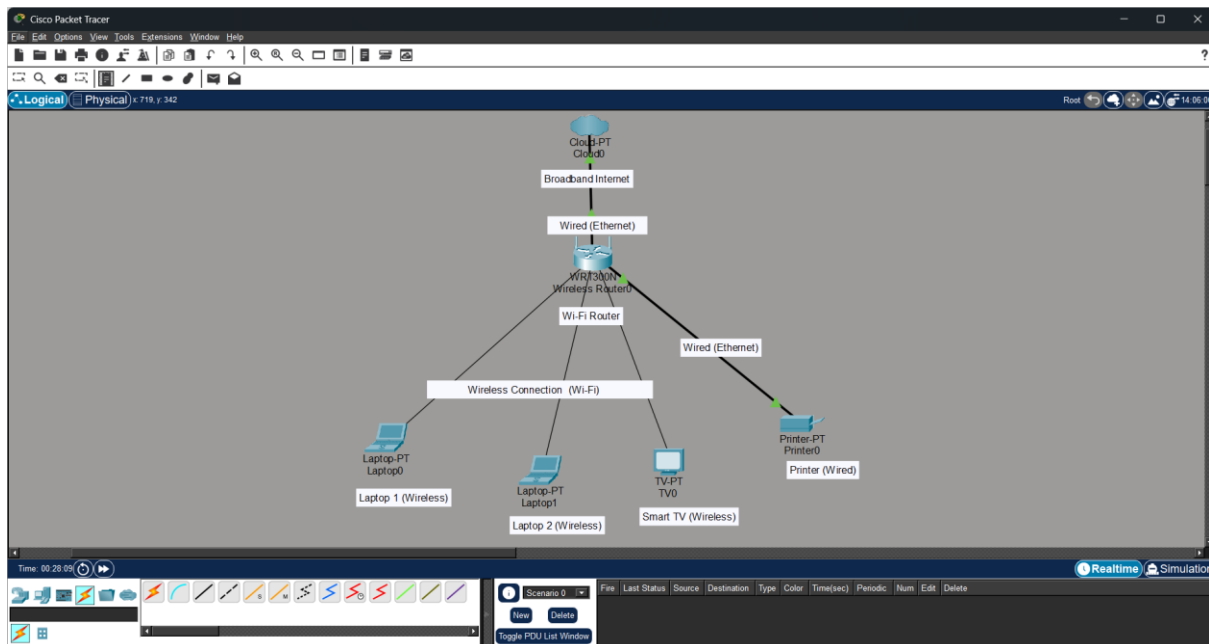
N+ - Network Security, Maintenance and Troubleshooting Procedures

Build a SOHO Network:

Q1. Draw a network diagram for a small office/home office (SOHO) setup that includes: one broadband internet connection, one Wi-Fi router, two laptops, one smart TV, and one printer. Clearly label each device and connection type (wired or wireless).

Ans:

Figure 1: SOHO Network Diagram in Cisco Packet Tracer



Connection Details:

From	To	Connection
Broadband Internet	Wi-Fi Router	Wired (Ethernet)
Wi-Fi Router	Laptop 1	Wireless (Wi-Fi)
Wi-Fi Router	Laptop 2	Wireless (Wi-Fi)
Wi-Fi Router	Smart TV	Wireless (Wi-Fi)
Wi-Fi Router	Printer	Wired (Ethernet)

Q2. List all the hardware components (with brand/model suggestions) you would need to physically build the SOHO network diagrammed above, including cables and any switches or extenders if needed.
Hint: Check current listings on Flipkart or Amazon for typical device models used in Indian homes/offices.

Ans:

Sr. No.	Hardware Component	Suggested Brand / Model	Quantity	Purpose
1	Broadband Internet Connection	JioFiber / Airtel Xstream Fiber	1	Provides Internet connectivity
2	Wi-Fi Router	TP-Link Archer C6 AC1200	1	Connects all wired and wireless devices
3	Laptop	Dell Inspiron 15	2	User devices for accessing the network
4	Smart TV	Samsung Smart TV 43"	1	Accesses online streaming services
5	Network Printer	HP LaserJet MFP M126nw	1	Prints documents over the network
6	Cat6 Ethernet Cable	D-Link Cat6 UTP Cable	2	Connects wired devices
7	RJ45 Connectors	D-Link Cat6 RJ45 Connector	4	Terminates Ethernet cables
8	Network Switch (Optional)	TP-Link TL-SG105	1	Adds more LAN ports if required
9	Wi-Fi Range Extender (Optional)	TP-Link RE205 AC750	1	Extends Wi-Fi coverage

Steps to Physically Build the SOHO Network:

1. Arrange all the required hardware components in a suitable location.
 2. Place the broadband modem and Wi-Fi router in a central location for better network coverage.
 3. Connect the Broadband Internet connection to the **WAN (Internet)** port of the Wi-Fi Router using a **Cat6 Ethernet cable**.
 4. Connect the Printer to one of the **LAN** ports of the Wi-Fi Router using a **Cat6 Ethernet cable with RJ45 connectors**.
 5. Place both Laptops and the Smart TV within the Wi-Fi coverage area of the router.
 6. Connect all devices to a power source and switch them **ON**.
 7. Verify that all physical connections are secure before starting the network configuration.
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Q3. Write step-by-step instructions to configure the Wi-Fi router for your SOHO network, including setting up a unique SSID, enabling WPA2 security, and changing the default admin password.

Ans: Steps to Configure the Wi-Fi Router:

Step 1: Connect the Router

1. Connect the broadband Internet cable to the **WAN (Internet)** port of the Wi-Fi router.
2. Connect the router to a power source and switch it **ON**.
3. Connect a laptop to the router using Wi-Fi or an Ethernet cable.

Step 2: Access the Router Configuration Page

1. Open a web browser.
2. Enter the router's default IP address (e.g., **192.168.1.1**) in the address bar.
3. Log in using the default administrator username and password.

Step 3: Configure a Unique SSID

1. Open the **Wireless Settings** page.
2. Locate the **SSID (Network Name)** field.

3. Enter a unique network name, for example **SOHO_Network_2026**.
4. Click **Save** or **Apply**.

Step 4: Enable WPA2 Security

1. Open the **Wireless Security** settings.
2. Select **WPA2-Personal (AES)** as the security mode.
3. Enter a strong Wi-Fi password, for example **SOHO@2026**.
4. Click **Save** or **Apply**.

Step 5: Change the Default Administrator Password

1. Open the **Administration** or **Management** settings.
2. Select **Change Administrator Password**.
3. Enter the current administrator password.
4. Enter a new strong administrator password, for example **Admin@2026**.
5. Confirm the new password and click **Save**.

Step 6: Verify the Configuration

1. Restart the Wi-Fi router if required.
2. Connect all devices using the new SSID and Wi-Fi password.
3. Verify that all devices can successfully access the Internet.

Result

The Wi-Fi router has been successfully configured with a unique SSID, **WPA2-Personal** security, and a new administrator password, providing secure wireless connectivity for the SOHO network.

Q4. Configure IP addresses for all devices in your SOHO network: assign static IPs to the printer and smart TV, and use DHCP for the laptops. Show a table listing each device, its MAC address (make one up), and the assigned IP address.

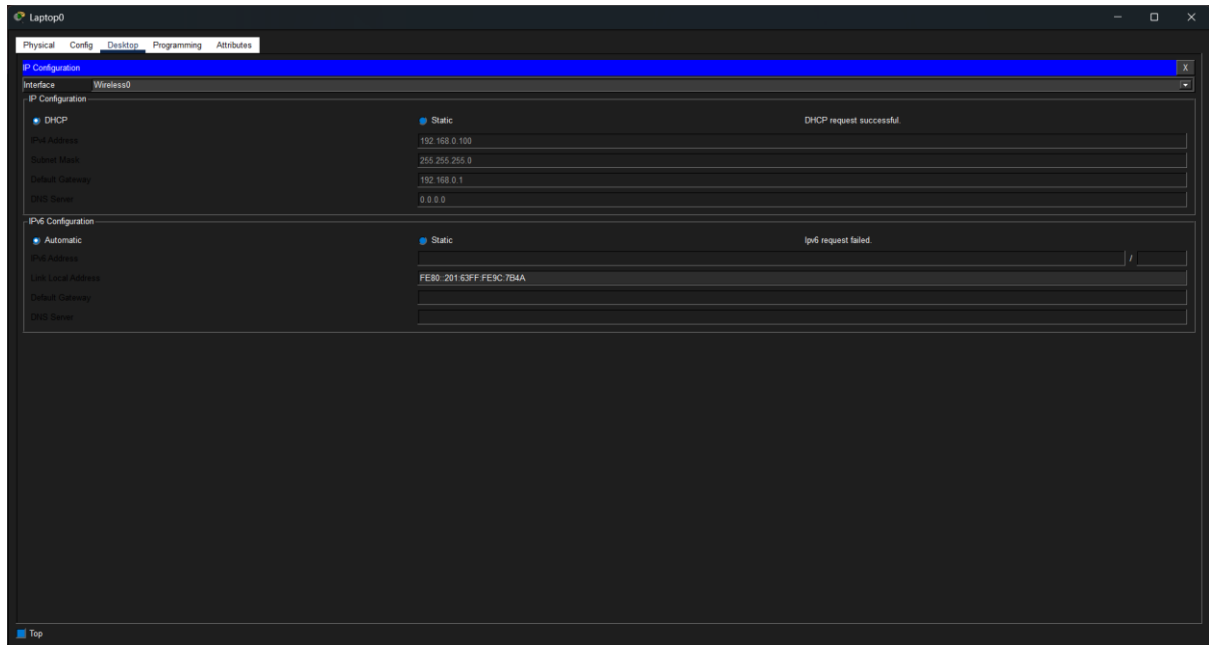
Ans: Steps to Configure IP Addresses:

Step 1: Configure Laptop 1 using DHCP

1. Open **Laptop 1**.
2. Go to **Desktop → IP Configuration**.

3. Select **DHCP**.
4. The IP address is automatically assigned by the router.

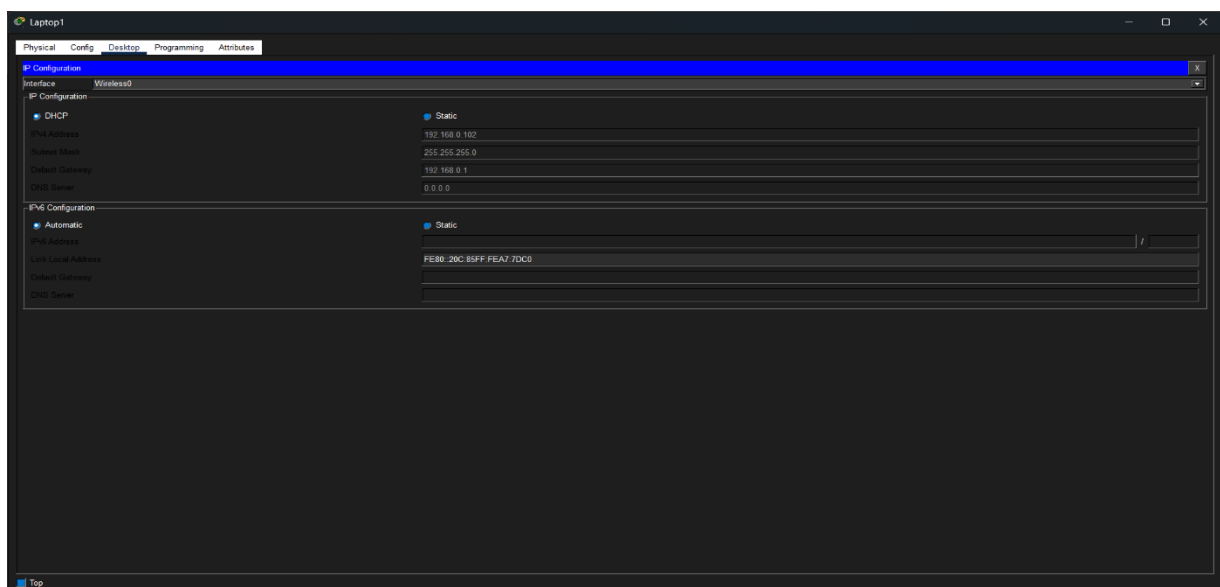
Figure 1: Laptop 1 IP Configuration using DHCP



Step 2: Configure Laptop 2 using DHCP

1. Open **Laptop 2**.
2. Go to **Desktop** → **IP Configuration**.
3. Select **DHCP**.
4. The IP address is automatically assigned by the router.

Figure 2: Laptop 2 IP Configuration using DHCP



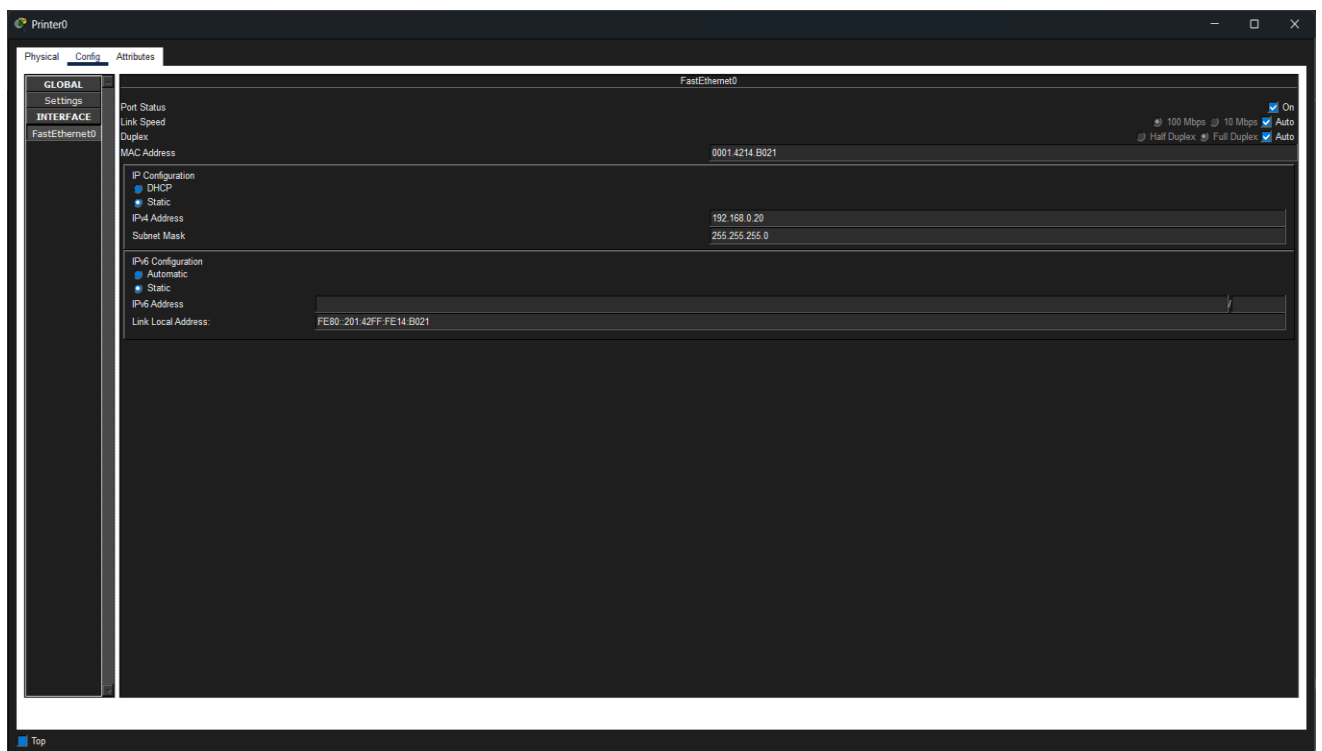
Step 3: Configure Smart TV with Static IP

1. Open the **Smart TV**.
2. Go to **Config → FastEthernet0**.
3. Select **Static** IP configuration.
4. Configure the following:
 - IP Address: **192.168.0.10**
 - Subnet Mask: **255.255.255.0**

Step 4: Configure Printer with Static IP

1. Open the **Printer**.
2. Go to **Config → FastEthernet0**.
3. Select **Static** IP configuration.
4. Configure the following:
 - IP Address: **192.168.0.20**
 - Subnet Mask: **255.255.255.0**

Figure 3: Printer Static IP Configuration



Device IP Address Table

Device	IP Assignment	MAC Address	Assigned IP Address
Laptop 1	DHCP	00:1A:2B:3C:4D:01	192.168.0.100
Laptop 2	DHCP	00:1A:2B:3C:4D:02	192.168.0.102
Smart TV	Static	00:1A:2B:3C:4D:03	192.168.0.10
Printer	Static	00:1A:2B:3C:4D:04	192.168.0.20

Q5. Explain how you would troubleshoot if one of the laptops cannot connect to the internet in your SOHO network. List at least 3 possible causes and what steps you would take to diagnose and fix each issue.

Ans: If one of the laptops cannot connect to the Internet, the following troubleshooting steps can be used to identify and resolve the problem.

Cause 1: Incorrect Wi-Fi Connection

Diagnosis

- Check whether the laptop is connected to the correct SSID (Wi-Fi network).
- Verify that the correct Wi-Fi password has been entered.

Solution

1. Reconnect the laptop to the correct SSID.
2. Enter the correct WPA2 Wi-Fi password.
3. If required, disconnect and reconnect to the wireless network.

Cause 2: IP Address Configuration Problem

Diagnosis

- Open Command Prompt and run the ipconfig command.
- Verify whether the laptop has received a valid IP address from the DHCP server.

Solution

1. Open IP Configuration and select DHCP.
2. Run ipconfig /release followed by ipconfig /renew.

3. Ensure that the router's DHCP server is enabled.
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Cause 3: Router or Internet Connection Failure

Diagnosis

- Check whether the router is powered ON.
- Verify that the broadband cable is connected to the WAN (Internet) port.
- Check whether other devices can access the Internet.

Solution

1. Restart the router.
 2. Check all cable connections.
 3. If all devices are offline, contact the Internet Service Provider (ISP).
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Cause 4: DNS Configuration Issue

Diagnosis

- Check whether the laptop is connected to the network but websites are not opening.
- Run the ping command to test network connectivity.

Solution

1. Configure a valid DNS server (e.g., 8.8.8.8 or 8.8.4.4).
 2. Restart the network connection.
 3. Test Internet access again.
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Result

The Internet connectivity issue was successfully diagnosed and resolved by checking the Wi-Fi connection, IP address configuration, router status, and DNS settings.